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RESEARCH INTERESTS

My research focuses on the control of renewable energy systems, viewed as large-scale, complex networked systems. I aim to develop theoretical methods that ensure stability, dynamic performance, and optimal operation of future large-scale energy systems by integrating control, optimization, and learning.

- **Stability and Dynamics:** Modeling, small-/large-signal stability analysis, and control of inverter-dominated energy systems, with emphasis on interactions and scalability
- **Coordination and Performance:** Coordinated and distributed control of large-scale inverter-based resources for optimized dynamic performance and ancillary services
- **Standardization:** Development of IEC-standardized models of inverter-based resources and next-generation grid-integration specifications (under IEC SC 8A)

APPOINTMENTS

- **Research Assistant Professor** Jul. 2025 – Present
Department of Automation, Tsinghua University, China
- **Senior Researcher** Nov. 2024 – Jan. 2025
Automatic Control Laboratory (IfA), ETH Zürich, Switzerland
- **Postdoctoral Researcher** Nov. 2021 – Oct. 2024
Automatic Control Laboratory (IfA), ETH Zürich, Switzerland
Advisor: Prof. Florian Dörfler

EDUCATION

- **Ph.D. in Control Science and Engineering** Aug. 2016 – Jun. 2021
Department of Automation, Tsinghua University, China
Dissertation: Research on Synchronization Issues in Power Systems with High Penetration of Power Electronics [\[link\]](#)
Advisor: Prof. Hua Geng
- **B.S. in Automation** Aug. 2012 – Jun. 2016
Department of Automation, Tsinghua University, China

AWARDS AND RECOGNITIONS

- **The 2024 ISGAN Award of Excellence RUNNER-UP** Mar. 2025
- **First Prize of Science and Technology Progress Award of China Electrotechnical Society (CES)** Dec. 2024
- **First Prize of Science and Technology Progress Award of China Power Supply Society (CPSS)** Nov. 2023
- **Stanford/Elsevier Top 2% Scientists** 2023, 2024, 2025

• Outstanding Doctoral Dissertation Nomination of CPSS	Apr. 2023
• Outstanding Doctoral Dissertation of Tsinghua University	Jun. 2021
• Beijing Outstanding Graduates	Jun. 2021
• Best Paper of IEEE Transactions on Energy Conversion	Dec. 2022
• Best Paper of High Voltage	Dec. 2023
• First Prize of the Basil Papadrias Award of IEEE PowerTech 2025	Jul. 2025
• Excellent Paper of IEEE PEAC 2018	Nov. 2018
• Excellent Paper of the 22nd CPSS Annual Conference	Nov. 2017
• Star Reviewer of IEEE Transactions on Energy Conversion	Dec. 2021
• Outstanding Reviewer of IEEE Transactions on Sustainable Energy	Feb. 2019
• The Second Prize, the First Beijing's Universities AI Academic Forum Award	Mar. 2019
• The First Class Fellowship, Tsinghua Graduate Student Professional Practice	Dec. 2018
• China National Scholarship	Dec. 2019
• China National Scholarship	Nov. 2018
• Best Presenter at IEEE PEAC 2018	Nov. 2018
• Best Presenter at the 22nd CPSS Annual Conference	Nov. 2017
• The Third Prize, Tsinghua University Student Laboratory Contribution Award	Mar. 2017
• Tsinghua University Academic Scholarship	Dec. 2013
• HAGE Scholarship	May 2013

TEACHING

- **Principles and Systems of Electrical Energy Conversion**, I served as one of two lecturers and delivered 46 h lectures among 64 h, Fall Semester 2025, Tsinghua University.
- **CPSS Power Electronics School**, I delivered 2 h lectures, Nov. 2025, China Power Supply Society.
- **Control Systems** (Prof. Florian Dörfler), Teaching Assistant, Fall Semester 2022, Spring Semester 2023, Fall Semester 2023, Spring Semester 2024, ETH Zürich.
- **Control Systems I** (Prof. Florian Dörfler), Teaching Assistant, Spring Semester 2022, ETH Zürich.
- **Fundamentals of Analog Electronics Technology** (Prof. Hua GENG), Teaching Assistant, for three semesters, Tsinghua University.
- **Electric Drive and Motion Control** (Prof. Geng YANG), Teaching Assistant, for two semesters, Tsinghua University.

STUDENT SUPERVISION

PhD Students

- **Verena Häberle (ETH)**: Dynamic ancillary services provision in future power systems, co-supervised with Prof. Florian Dörfler, 2022 – 2025, wins the Hans-Eggenberger-Award 2025.

- **Xiang Zhu (Tsinghua):** Optimization of frequency response characteristics of renewable energy systems, co-supervised with Prof. Hua Geng, Jan. 2025 – present.
- **Shanshan Jiang (Tsinghua):** Stability control of DC microgrids during mode switching, co-supervised with Prof. Hua Geng, Jan. 2025 – present.
- **Zelin Sun (Tsinghua):** Decentralized stability analysis and control for AC/DC power systems, co-supervised with Prof. Hua Geng, Jan. 2025 – present.

Master Students

- **Roger Domingo-Enrich (ETH):** Complex frequency control of dynamic virtual power plants, Spring 2023.
- **Ramona Stoll (ETH):** Grid code validation of cross-forming controlled converters under current limitation, Fall 2024.
- **Martin Pedersen (ETH):** Aggregated modeling of grid-forming converters with complex droop control, Spring 2023.
- **Maitraya Avadhut Desai (ETH):** Stable current-limiting control for grid-forming converters, Fall 2022.

RESEARCH PROJECTS

- **National Natural Science Foundation of China (NSFC), Excellent Young Scientists Fund Overseas Program:** Renewable Energy Integration 2026 - 2028
Principal Investigator (CNY 1 to 3 million)
- **Horizon Europe AGISTIN Project:** Advanced Grid Interfaces for innovative STorage INtegration (AGISTIN) [\[link\]](#) Jan. 2023 – Jan. 2025
T3.5 Task Leader
- **Horizon Europe POSYTYF Project:** POWering SYstem flexibiliTY in the Future (POSYTYF) through Renewable Energy Sources [\[link\]](#) Nov. 2021 – May 2024
T4.2 Task Leader
- **IEC TS 63406:2025:** Generic RMS Simulation Models of Converter-Based Generating Units for Power System Dynamic Analysis [\[link\]](#) Apr. 2018 – Oct. 2025
Principal Drafter; Secretary of IEC SC 8A WG 8
- **Young Innovation Foundation of Beijing National Research Center for Information Science and Technology:** Distributed Control Theories and Technologies for Lunar Base Energy Systems Jan. 2026 – Dec. 2027
Principal Investigator (CNY 0.3 million)
- **Dongfang Electric Autocontrol Engineering Co., Ltd.:** Research and Development of Key Technologies for Full-Condition GFM PCSs Jan. 2026 – Dec. 2026
Principal Investigator (CNY 1.43 million)
- **CSG Electric Power Research Institute:** Standardized Modeling of Wind-Solar-Storage Complementary Systems and Grid-Forming Control Dec. 2025 – Mar. 2027
Co-Principal Investigator (CNY 1.35 million)
- **National Natural Science Foundation of China:** Flexible Grid-Connection Control and Application of Wind Energy Conversion Systems Jan. 2018 – Dec. 2020
Key Participant

- **National Natural Science Foundation of China:** Large-Disturbance Stability and Cascading Failure in Power Systems with High Penetration of Power Electronics Jan. 2021 – Dec. 2024
Key Participant

ACADEMIC SERVICES

- **IEC Standardization Activities:** Expert, WG Secretary of IEC SC 8A WG 8 *Modeling of Renewable Energy Generation for Power System Dynamic Analysis*
- **Session Chair:** IECON 2021, RPG 2021, FAFEE 2026
- **Invited Session Speeches:** IECON 2021, SPIES 2022, PEET Workshop 2023, FAFEE 2025, EI² 2025, Champéry Power Conference 2026
- **Guest Editor:** Cybernetics and Intelligence, Journal of Electrical Engineering
- **Workshop Organizer** Power Electronics Emerging Technologies (PEET) Workshop 2025 (Xi'an)
- **Committee Member:** Artificial Intelligence Technical Committee of CPSS
- **Membership:** IEEE, CPSS, CAA
- **Reviewer**

Power Systems Journals: IEEE TPWRS / TSTE / TSG / TEC / TPWRD / OAJPE / PESL / PEM, CSEE JPES, IJEPES, IET GTD / ESI

Power Electronics Journals: IEEE TPEL / TIE / JESTPE

Control Journals: IEEE TCNS, Automatica, Control Engineering Practice

Interdisciplinary Journals: IEEE TIA / TII / JETCAS / Access

Received the Star Reviewer Award of IEEE TEC (2021) and the Outstanding Reviewer Award of IEEE TSTE (2019).

TALKS

- **Invited Tutorial at IECON 2021:** *Transient stability of power electronics-dominated power networks: principle and application*, Oct. 2021, remote. [\[pdf\]](#)
- **Invited Talk at the Fourth Champéry Power Conference:** *Quantifying stability conditions for grid-forming and grid-following heterogeneous power systems*, Feb. 2026, Champéry, Switzerland. [\[pdf\]](#)
- **Plenary Speech at the Power Electronics Emerging Technologies Workshop:** *Multivariable grid-forming control and nonlinear stability analysis*, Oct. 2023, Nanjing, China. [\[pdf\]](#)
- **Invited Lecture at the CPSS Advanced Technologies School:** *Transient operation and stability control of grid-forming converters*, Nov. 2025, Shanghai, China. [\[pdf\]](#)
- **Invited Session Talk at EI² 2025:** *Lunar surface power grid architecture and control*, Dec. 2025, Jilin, China.
- **Invited Session Talk at FAFEE 2025:** *Complex-frequency grid-forming control*, May 2025, Xiamen, China. [\[pdf\]](#)
- **Invited Tutorial at EirGrid:** *Grid-forming, grid-following, and cross-forming controls*, Aug. 2024, remote. [\[pdf\]](#)
- **Invited Talk at Hefei University of Technology:** *Grid-forming control and decentralized stability analysis of renewable energy systems*, Jan. 2026, Hefei, China. [\[pdf\]](#)

- **Invited Session Talk at International Conference on Smart Power & Internet Energy Systems (SPIES 2022):** *Complex-frequency synchronization and multivariable stability analysis*, Dec. 2022, remote.
- Oral Speech at IECON 2024, *Dynamic complex-frequency control of grid-forming converters*, Nov. 2024, Chicago, USA.
- Oral Speech at the Third Champéry Power Conference, *Stability guarantees for grid-forming complex droop control*, Feb. 2024, Champéry, Switzerland.
- Coffee Talk at IfA, ETH, *Multivariable stability of converter-based power systems: centralized and decentralized methods*, Apr. 2023, ETH Zürich.
- Oral Speech at IAS Annual Meeting 2020, *Synchronization stability analysis and enhancement of grid-tied multi-converter systems*, Oct. 2020, remote.
- Oral Speech at PESGM 2020, *Transient stability analysis and enhancement of renewable energy conversion system during LVRT*, Aug. 2020, remote.
- Oral Speech at IEC SC 8A ahG 3 Meeting, *Modeling of inverter-based generation for power system stability studies*, Oct. 2019, Nanjing, China.
- Oral Speech at PESGM 2019, *Modeling recommendation of wind turbine/farm for power system stability studies*, Aug. 2019, Atlanta, USA.
- Oral Speech at PESGM 2019 (on behalf of Prof. GENG), *Hardware-in-the-loop (HIL) tools for power engineering education*, Aug. 2019, Atlanta, USA.
- Oral Speech at PESGM 2019, *Dynamic equivalent modeling of wind power plants for various timescale small signal stability analyses*, Aug. 2019, Atlanta, USA.
- Oral Speech at IEEE PEAC 2018, *Synchronization stability analysis of grid-tied power converters under severe grid voltage sags*, Nov. 2018, Shenzhen, China. (Best Presenter Award)

PUBLICATIONS

Topic I: Stability and Dynamics in Inverters-Dominated Power Systems

- [1] **X. He**, V. Häberle, and F. Dörfler, “Complex-frequency synchronization of converter-based power systems,” *IEEE Trans. Control Netw. Syst.*, vol. 12, no. 1, pp. 787–799, 2025, [\[link\]](#)
- [2] **X. He**, M. A. Desai, L. Huang, and F. Dörfler, “Cross-forming control and fault current limiting for grid-forming inverters,” *IEEE Trans. Power Electron.*, vol. 40, no. 3, pp. 3980–4007, 2025, [\[link\]](#)
- [3] **X. He**, L. Huang, I. Subotić, V. Häberle, and F. Dörfler, “Quantitative stability conditions for grid-forming converters with complex droop control,” *IEEE Trans. Power Electron.*, vol. 39, no. 9, pp. 10 834–10 852, 2024, [\[link\]](#)
- [4] **X. He** and F. Dörfler, “Passivity and decentralized stability conditions for grid-forming converters,” *IEEE Trans. Power Syst.*, vol. 39, no. 3, pp. 5447–5450, 2024, [\[link\]](#)
- [5] **X. He**, C. He, S. Pan, H. Geng, and F. Liu, “Synchronization instability of inverter-based generation during asymmetrical grid faults,” *IEEE Trans. Power Syst.*, vol. 37, no. 2, pp. 1018–1031, 2022, [\[link\]](#)
- [6] L. Huang[#], **X. He[#]** (Co-first author), H. Xin, Z. Li, P. Ju, F. Ma, K. Wang, and F. Dörfler, “System strength in power electronics-dominated power systems: An enabler to stability and control,” *IEEE Power Energy Mag.*, vol. 24, no. 1, pp. 49–66, 2026, [\[link\]](#)

- [7] M. A. Desai, **X. He** (Corresponding author), L. Huang, and F. Dörfler, “Saturation-informed current-limiting control for grid-forming converters,” *Electr. Power Syst. Res.*, vol. 234, p. 110746, 2024, [\[link\]](#)
- [8] **X. He**, V. Häberle, I. Subotić, and F. Dörfler, “Nonlinear stability of complex droop control in converter-based power systems,” *IEEE Control Syst. Lett.*, vol. 7, pp. 1327–1332, 2023, [\[link\]](#)
- [9] C. He, **X. He**, H. Geng, H. Sun, and S. Xu, “Transient stability of low-inertia power systems with inverter-based generation,” *IEEE Trans. Energy Convers.*, vol. 37, no. 4, pp. 2903–2912, 2022, [\[link\]](#) (Best Paper for the period 2021–2022)
- [10] **X. He**, S. Pan, and H. Geng, “Transient stability of hybrid power systems dominated by different types of grid-forming devices,” *IEEE Trans. Energy Convers.*, vol. 37, no. 2, pp. 868–879, 2022, [\[link\]](#)
- [11] **X. He** and H. Geng, “PLL synchronization stability of grid-connected multiconverter systems,” *IEEE Trans. Ind. Appl.*, vol. 58, no. 1, pp. 830–842, 2022, [\[link\]](#)
- [12] **X. He**, H. Geng, J. Xi, and J. M. Guerrero, “Resynchronization analysis and improvement of grid-connected VSCs during grid faults,” *IEEE J. Emerg. Sel. Top. Power Electron.*, vol. 9, no. 1, pp. 438–450, 2021, [\[link\]](#)
- [13] **X. He** and H. Geng, “Transient stability of power systems integrated with inverter-based generation,” *IEEE Trans. Power Syst.*, vol. 36, no. 1, pp. 553–556, 2021, [\[link\]](#)
- [14] **X. He**, H. Geng, R. Li, and B. C. Pal, “Transient stability analysis and enhancement of renewable energy conversion system during LVRT,” *IEEE Trans. Sustain. Energy*, vol. 11, no. 3, pp. 1612–1623, 2020, [\[link\]](#)
- [15] **X. He**, H. Geng, and G. Mu, “Modeling of wind turbine generators for power system stability studies: A review,” *Renew. Sust. Energ. Rev.*, vol. 143, p. 110865, 2021, [\[link\]](#)
- [16] **X. He**, H. Geng, and S. Ma, “Transient stability analysis of grid-tied converters considering PLL’s nonlinearity,” *CPSS Trans. Power Electron. Appl.*, vol. 4, no. 1, pp. 40–49, 2019, [\[link\]](#)
- [17] **X. He**, H. Geng, and G. Yang, “Reinvestigation of single-phase FLLs,” *IEEE Access*, vol. 7, pp. 13 178–13 188, 2019, [\[link\]](#)
- [18] **X. He**, H. Geng, and G. Yang, “A generalized design framework of notch filter based frequency-locked loop for three-phase grid voltage,” *IEEE Trans. Ind. Electron.*, vol. 65, no. 9, pp. 7072–7084, 2018, [\[link\]](#)
- [19] H. Geng, C. He, Y. Liu, **X. He**, and M. Li, “Overview on transient synchronization stability of renewable-rich power systems,” *High Voltage Engineering*, vol. 48, no. 9, pp. 3367–3383, 2022, [\[link\]](#) (Best Paper Award, In Chinese)
- [20] V. Häberle, **X. He**, L. Huang, F. Dörfler, and S. Low, “Decentralized parametric stability certificates for grid-forming converter control,” *IEEE Trans. Control Netw. Syst.*, 2026, accepted, [\[link\]](#)
- [21] S. Jiang, Z. Sun, J. Zhang, M. Zhang, C. Tang, **X. He*** (Corresponding author), and H. Geng*, “A graph-theoretic approach for ensuring stability in DC microgrid networks during mode switching,” *IEEE Trans. Smart Grid*, vol. 17, no. 2, pp. 922–935, 2026, [\[link\]](#)
- [22] Z. Sun, S. Jiang, J. Zhang, **X. He**, and H. Geng, “Decentralized stability analysis for DC microgrids with heterogeneous dynamics,” *IEEE Trans. Smart Grid*, pp. 1–13, 2025, early access, [\[link\]](#)
- [23] M. Abdalmoaty, V. Häberle, **X. He**, and F. Dörfler, “Ultrafast grid impedance identification in dq -asymmetric three-phase power systems,” *Electr. Power Syst. Res.*, 2026, accepted, [\[link\]](#)

- [24] X. Zhu, **X. He**, H. Qing, and H. Geng, “Decentralized analysis approach for oscillation damping in grid-forming and grid-following heterogeneous power systems,” *arXiv preprint arXiv:2601.21562*, 2026, [\[link\]](#)
- [25] J. Miller, M. A. Desai, **X. He**, R. S. Smith, and G. Hug, “Network-independent incremental passivity conditions for grid-forming inverter control,” *arXiv preprint arXiv: 2506.14469*, 2025, [\[link\]](#)
- [26] K. Zhuang, L. Huang, H. Xin, **X. He**, V. Häberle, and F. Dörfler, “Dispatchable current source virtual oscillator control achieving global stability,” *arXiv preprint arXiv:2510.26977*, 2025, [\[link\]](#)
- [27] K. Zhuang, H. Xin, V. Häberle, **X. He**, L. Huang, and F. Dörfler, “Quantifying grid-forming behavior: Bridging device-level dynamics and system-level stability,” *arXiv preprint 2503.24152*, 2025, [\[link\]](#)
- [28] M. A. Desai, R. Stoll, **X. He**, F. Dörfler, and G. Hug, “Comparison of current-limiting strategies for grid-forming converters under grid faults,” in *Proc. 2025 IEEE Kiel PowerTech*, 2025, pp. 1–6, [\[link\]](#) (Best Paper Award)
- [29] H. Liu, **X. He**, H. Qing, Y. Chen, and H. Geng, “Stability-constrained GFM/GFL capacity ratio: Modeling and evaluation,” in *Proc. 2025 IEEE Industry Applications Society Annual Meeting (IAS)*, 2025, pp. 1–6, [\[link\]](#)
- [30] Z. Sun, S. Jiang, J. Zhang, M. Zhang, **X. He**, and H. Geng, “Decentralized stability analysis and region of attraction estimation for multi-bus DC microgrids,” in *Proc. 2025 IEEE Industry Applications Society Annual Meeting (IAS)*, 2025, pp. 1–6, [\[link\]](#)
- [31] V. Häberle, L. Huang, **X. He**, R. S. Smith, and F. Dörfler, “MIMO grid impedance identification of three-phase power systems: Parametric vs. nonparametric approaches,” in *Proc. 62nd IEEE Conference on Decision and Control (CDC)*, 2023, [\[link\]](#)
- [32] C. He, **X. He**, and H. Geng, “Transient stability of low-inertia power grid with inverter-based generations,” in *Proc. 10th Renewable Power Generation Conference (RPG 2021)*, 2021, pp. 91–97, [\[link\]](#)
- [33] J. Li, **X. He**, and H. Geng, “Grid code formulation recommendations on inverter-based generation during asymmetrical grid faults,” in *Proc. 10th Renewable Power Generation Conference (RPG 2021)*, 2021, pp. 289–295, [\[link\]](#)
- [34] **X. He** and H. Geng, “Synchronization stability analysis and enhancement of grid-tied multi-converter systems,” in *Proc. 2020 IEEE Industry Applications Society Annual Meeting*, 2020, pp. 1–8, [\[link\]](#)
- [35] **X. He**, H. Geng, and G. Yang, “Synchronization stability analysis of grid-tied power converters under severe grid voltage sags,” in *Proc. 2018 IEEE International Power Electronics and Application Conference and Exposition (PEAC)*, 2018, pp. 1–6, [\[link\]](#) (Excellent Paper Award)
- [36] X. Jiao, **X. He**, H. Geng, and B. Ren, “Hybrid average-value modelling of DFIG-based wind energy conversion systems,” in *Proc. 10th Renewable Power Generation Conference (RPG 2021)*, 2021, pp. 193–198, [\[link\]](#)
- [37] **X. He**, H. Geng, and G. Yang, “Mode clustering based dynamic equivalent modeling of wind farm for small-signal stability analysis,” *arXiv preprint arXiv:2109.08383*, 2021, [\[link\]](#)
- [38] **X. He**, H. Geng, and G. Yang, “Dynamic equivalent modeling of wind power plants for various timescale small signal stability analyses,” in *Proc. 2019 IEEE Power and Energy Society General Meeting (PESGM)*, 2019, pp. 1–5, [\[link\]](#)
- [39] **X. He** and H. Geng, “An overview on wind farm modelling for power system stability studies,” in *Proc. 8th Renewable Power Generation Conference (RPG 2019)*, 2019, pp. 1–8, [\[link\]](#)

- [40] **X. He**, H. Geng, G. Yang, X. Zou, and Y. Li, “Equivalent modelling of wind farm for small-signal stability analysis in weak power system,” *J. Eng.*, vol. 2017, no. 13, pp. 1388–1393, 2017, [\[link\]](#)

Topic II: Coordination and Dynamic Performance of Inverter-Based Resources

- [41] **X. He**, J. Duarte, V. Häberle, and F. Dörfler, “Aggregate grid-forming control of distributed energy resources,” *IEEE Trans. Smart Grid*, pp. 1–13, 2026, early access, [\[link\]](#)
- [42] **X. He**, G. Hua, Y. Geng, and Z. Xin, “Startup and integration control strategy of DFIG based isolated wind farm connected with LCC-HVDC,” *Autom. of Elec. Power Syst.*, vol. 43, no. 9, pp. 99–107, 2019, [\[link\]](#) (In Chinese)
- [43] **X. He**, H. Geng, G. Yang, and X. Zou, “Coordinated control for large-scale wind farms with LCC-HVDC integration,” *Energies*, vol. 11, no. 9, 2018, [\[link\]](#)
- [44] V. Häberle, **X. He**, L. Huang, E. Prieto-Araujo, and F. Dörfler, “Optimal dynamic ancillary services provision based on local power grid perception,” *IEEE Trans. Power Syst.*, vol. 40, no. 2, pp. 1816–1831, 2025, [\[link\]](#)
- [45] V. Häberle, A. Tayyebi, **X. He**, E. Prieto-Araujo, and F. Dörfler, “Grid-forming and spatially distributed control design of dynamic virtual power plants,” *IEEE Trans. Smart Grid*, vol. 15, no. 2, pp. 1761–1777, 2024, [\[link\]](#)
- [46] V. Häberle, L. Huang, **X. He**, E. Prieto-Araujo, and F. Dörfler, “Dynamic ancillary services: From grid codes to transfer function-based converter control,” *Electr. Power Syst. Res.*, vol. 234, p. 110760, 2024, [\[link\]](#)
- [47] C. Feng, L. Huang, **X. He**, Y. Wang, F. Dörfler, and C. Kang, “Hybrid oscillation damping and inertia management for distributed energy resources,” *IEEE Trans. Power Syst.*, vol. 40, no. 6, pp. 5041–5056, 2025, [\[link\]](#)
- [48] X. Zhu, H. Geng, H. Qing, G. Ruan, and **X. He**, “Dynamic virtual power plants with robust frequency regulation capability,” *IEEE Trans. Ind. Appl.*, vol. 62, no. 1, pp. 132–143, 2026, [\[link\]](#)
- [49] R. Domingo-Enrich, **X. He**, V. Häberle, and F. Dörfler, “Dynamic complex-frequency control of grid-forming converters,” in *Proc. 2024 Annual Conference of the IEEE Industrial Electronics Society (IECON)*, 2024, [\[link\]](#)
- [50] W. Schitteck, N. Wiese, C. Hachmann, **X. He**, J. Heid, J. Ferry, and M. Braun, “How to tailor power-electronic inertia to the constraints of its resource,” *TechRxiv preprint*, July 2025, [\[link\]](#)
- [51] K. Zhuang, H. Xin, H. Chen, **X. He**, H. Gao, and L. Huang, “On the evaluation of frequency-forming and voltage-forming capabilities of power converters,” in *Proc. 2025 IEEE Power and Energy Society General Meeting (PESGM)*, 2025, pp. 1–5, [\[link\]](#)
- [52] S. Zhao, **X. He**, C. Lyu, Y. Zhou, S. Xu, X. Liu, and H. Geng, “Black start-up and coordinated control strategy of standalone doubly-fed wind farms connected to LCC-HVDC,” *Power System and Clean Energy*, vol. 37, no. 7, pp. 87–96, 2021, [\[link\]](#) (In Chinese)
- [53] **X. He**, H. Geng, G. Yang, and X. Zou, “VSG control for DFIG-based islanded wind farm with LCC-HVDC integration,” in *Proc. 2018 IEEE Power and Energy Society General Meeting (PESGM)*, 2018, pp. 1–5, [\[link\]](#)
- [54] J. Xi, H. Geng, and **X. He**, “Adaptive VSG control scheme for large scale wind farms to improve frequency response characteristics,” in *Proc. 2019 IEEE Industry Applications Society Annual Meeting*, 2019, pp. 1–7, [\[link\]](#)

Topic III: Standardization in Inverters-Dominated Power Systems

- [55] V. Häberle, K. Zhuang, **X. He**, L. Huang, G. Hug, and F. Dörfler, “Next-generation grid codes: Toward a new paradigm for dynamic ancillary services,” *arXiv preprint arXiv:2601.07090*, 2026, [\[link\]](#)
- [56] **IEC SC 8A WG 8**, IEC TS 63406:2025: Generic RMS simulation models of converter-based generating units for power system dynamic analysis, Oct. 2025. [\[link\]](#)

PATENTS

- [1] **X. He**, M.A. Desai, L. Huang, F. Dörfler. Method and a controller for controlling a grid-forming converter. EP24169470.2 (PCT international patent), filed, Apr. 10, 2024.
- [2] **X. He**, Y. Liu, J. Yang, S. Yao. Method and apparatus for multimodal vibration suppression of a tower structure. CN121611568A, Mar. 6, 2026.
- [3] H. Geng, S. Jiang, Z. Sun, **X. He**. Graph-theory-based network mode switching control method and apparatus for DC microgrids. CN121663441A, Mar. 13, 2026.
- [4] X. Meng, H. Geng, C. He, **X. He**, G. Li, T. Zhou. Method and apparatus for transient frequency stability control of renewable energy grid-connected equipment. CN120749790A, Oct. 3, 2025.
- [5] H. Geng, S. Zhao, **X. He**, S. Xu, C. Lü, Y. Zhou. Multi-device coordinated reactive power support method and apparatus for suppressing transient overvoltage. ZL202111435905.8, Nov. 29, 2021.
- [6] H. Geng, **X. He**, S. Zhao. Method, electronic device, and storage medium for evaluating synchronization stability margin of converters. ZL202111194284.9, Oct. 13, 2021.
- [7] H. Geng, **X. He**, G. Yang. Phase-locked control method and unit for low-voltage ride-through, and transient stability control method and system. ZL201810634917.5, Jun. 20, 2018.
- [8] H. Geng, **X. He**, S. Guo. Electromagnetic torque calculation and feedback control method in I/F control of permanent magnet synchronous motors. ZL201811076802.5, Sep. 14, 2018.
- [9] H. Geng, **X. He**, G. Yang. Coordinated control method for offshore wind farms connected to conventional HVDC systems. ZL201810194157.0, Mar. 9, 2018.
- [10] X. Zou, J. Liu, H. Geng, **X. He**, D. Xia. Virtual synchronous control method and apparatus for islanded doubly-fed wind farms connected to HVDC systems. ZL201810087695.X, Jan. 30, 2018.

Updated in Mar. 2026.